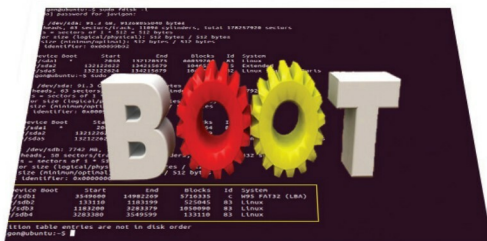


Resolving Boot Loader Issues

Boot loader issues often plague computer systems. This article demystifies the booting process, both for BIOS based systems as well as UEFI systems.



A Basic Input Output System (BIOS) powered system goes through a number of steps from when it is powered up to when it runs a Red Hat Enterprise Linux (RHEL) system. The following list of steps gives a high-level overview of what exactly takes place. This list also applies to virtual machines that emulate a traditional BIOS system. The list assumes that the GRUB2 boot loader is being used. For different boot loaders, the list will differ.

- 1) The BIOS firmware is started and performs a 'power on self-test' (POST).
- 2) The BIOS scans for possible boot devices, and orders them according to a user-set preference.
- 3) The boot devices are scanned for boot firmware (such as a PXE ROM on network cards), a Master Boot Record (MBR), or a partition marked as bootable. If found, the BIOS executes them.
- 4) The first-stage boot loader stored in the MBR loads the Stage 1.5 (drivers) and Stage 2 boot loaders from the disk, and executes them.
- 5) The boot loader loads a configuration file from the disk. In the case of GRUB2, this will be `/boot/grub2/grub.cfg`. The file is shown in Figure 1. This file is auto generated and should not normally be edited manually.
- 6) The configuration file is parsed and, based on its contents, a boot entry is selected automatically by the user.
- 7) The kernel and initial RAM disk referenced in the boot

entry are loaded from the disk, and control is handed over to the kernel.

- 8) The kernel starts and initialises hardware using the drivers found in the initial RAM disk. A simple init system is also started from the RAM disk.
- 9) The scripts in the initial RAM disk mount the root file system of the target system; they then switch the root to the newly mounted file system and hand over control to `/sbin/init` on the target root file system.
- 10) The init system mounts file systems and starts services according to its configuration.

GRUB2

The boot loader used on RHEL is GRUB2, the second major version of the GRand Unified Bootloader. GRUB2 stores its files on a BIOS system in a number of different locations, as listed below.

`/boot/`

The kernel and initial RAM disks are stored here. Sub-directories contain other files.

`/boot/grub2/`

This contains the configuration files, extension modules